

PYTHON INNER WORKING

python hello.py -> ye krte hi code convert ho jata hai byte code mein (mostly hidden) aur fir python VM mein code run ho jata hai

Step 1 -> compile to Byte Code

- Byte Code is low level code and platform independent i.e. bas ek python VM chahiye)
- Byte Code runs faster
- .pyc -> compiled python (also known as frozen binaries)
- __pycache__ -> ye python ka system folder hai
- hello.cpython-314.pyc (ye ek file bani hai __pycache__ mein) -> for source change and python version
- .pyc works only for imported files

PVM(Python VM)

- It is a code loop to iterate byte code(ya fir normal files bhi)
- Python is interpreted language (line by line chltta hai)
- PVM is a run time engine
- Byte Code is not machine code
- Machine code is direct instruction for hardware like assembly language
- Byte Code is python specific interpretation
- cpython is standard implementation
- jython for Java Binaries, IronPython, Stackless (for working with concurrencies) , PyPy(for performance oriented work)